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Position-specific differences in match performance of elite U-17 soccer players using GPS-derived external load data: focus on forwards and defenders

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[Purpose] The objective of this study was to investigate differences in match performance characteristics between forwards and defenders among elite U-17 soccer players using Global Positioning System (GPS)-derived data.

[Methods] This study included 37 elite U-17 soccer players representing Gyeonggi Province. External load data collected during official matches were analyzed. Variables were classified as measures of aerobic capacity, anaerobic capacity, neuromuscular performance, and agility and change-of-direction ability. Between-group differences were assessed using independent samples *t*-tests.

[Results] No significant positional differences were observed in total distance covered, distance per minute, mean speed, or low- to moderate-intensity activity zones ($p > 0.05$). Compared with defenders, forwards demonstrated significantly higher values for high-speed running distance, a higher proportion of high-intensity activity, maximum speed, and high-intensity acceleration and change-of-direction performance ($p < 0.05$).

[Conclusion] These findings suggest that at the elite youth level, fundamental aerobic capacity is similarly developed across playing positions, whereas high-intensity movement and neuromuscular performance capacities vary according to playing roles and tactical demands. Accordingly, training programs and player development strategies should be designed to account for positional characteristics.

[Keywords] U-17 soccer players, GPS, forward, defender, match performance characteristics

INTRODUCTION

Youth soccer players exhibit an intermittent exercise pattern characterized by repeated activities of varying intensities during match play, warranting a combination of both aerobic and anaerobic capacities. A soccer match typically lasts approximately 90 min, during which players cover distances of 9-12 km while repeatedly performing high-intensity sprinting, acceleration-deceleration, and changes in direction[1]. These characteristics necessitate both aerobic endurance to sustain match performance and anaerobic capacity to execute sprints and high-intensity movements, which are key determinants of overall performance[2].

Running demands are not limited to the total distance covered but also include multiple indicators such as high-intensity running, sprinting, and acceleration-deceleration activities, all of which are closely associated with the physical fitness levels of players[3]. Adolescence is a period of simultaneous growth and development during which physical capacities such as speed, agility, and aerobic fitness play critical roles in performance and exhibit developmental patterns distinct from those of adults[4]. Therefore, accurately understanding match performance among young soccer players requires comprehensive consideration of various physical fitness components and analysis of position-specific physical characteristics.

Notably, positional roles in soccer impose distinct physical and physiological demands. Defenders are frequently exposed to situations involving opponent containment, aerial duels, and physical contact, making body size, muscular strength, and anaerobic performance particularly important[5,6]. In youth players, biological maturation and body size influence positional assignment, with central defenders tending to be taller and heavier[6,7]. In contrast, forwards are required to penetrate confined spaces and execute goal-scoring actions, highlighting the importance of high-intensity performance abilities such as sprinting, jumping, and explosive power[8]. Indeed, forwards tend to demonstrate superior performance in short-distance sprinting and jumping compared with defenders[8]. These differences translate into variations in high-intensity activity frequency and movement patterns during matches[3]. Accordingly, analyzing dif-

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ferences in physical characteristics between forwards and defenders provides critical evidence for designing position-specific training programs and informing player selection and development strategies.

With advancements in sports science, Global Positioning System (GPS)-based analysis has emerged as a key tool for quantifying player movement during match play. GPS systems provide real-time data on multiple external load indicators, including total distance, high-speed running (HSR), sprinting, acceleration-deceleration, and maximum speed, enabling an objective evaluation of the physical demands of competition[9,10]. In young players, whose physical capacities fluctuate considerably during growth, GPS-based analysis plays an important role in assessing both developmental status and match demands[6]. Furthermore, GPS is effective in identifying position-specific activity patterns and differences in physical demands, with considerable variations observed in both total distance and high-intensity activity across positions[3,11]. Therefore, position-specific analysis using GPS data, particularly comparisons between forwards and defenders, is essential for scientific training design and player development strategies.

In this study, we aimed to analyze the physical characteristics of forwards and defenders among youth soccer players using GPS data. To achieve this, external load indicators, including total distance, high-intensity running, sprinting, and acceleration-deceleration during matches, were compared and analyzed to identify differences between the two positions.

METHODS

Participants

This study included 37 elite U-17 soccer players from Gyeonggi Province. All participants were elite players registered with the Korea Football Association and were actively involved in regular team training and official matches. Participants were selected if they had been injury-free for at least six months and were able to fully participate in both training and competition. According to their match positions, players were categorized into a defender group



Figure 1. Flowchart of the study design.

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Table 1. Characteristics of Participants

Group	Years (age)	Height (cm)	Weight (kg)	BMI (kg/m²)
DG (*n* = 17)	16.65 ± 0.49	179.35 ± 5.97	69.76 ± 6.19	21.67 ± 2.40
FG (*n* = 20)	16.60 ± 0.50	178.55 ± 5.86	69.15 ± 5.19	21.69 ± 2.16

Values are mean ± SD, defender group = DG, forward group = FG, body mass index = BMI.

Table 2. Differences in Aerobic Capacity

Variables	DG (*n* = 17)	FG (*n* = 20)	*t*	*p*	*d*
Total distance (km)	3.99 ± 0.37	4.08 ± 0.58	-0.573	0.570	-0.19
DPM (m/min)	103.64 ± 8.09	106.38 ± 7.61	-1.060	0.296	-0.35
AVE speed (km/h)	6.20 ± 0.47	6.35 ± 0.44	-0.996	0.326	-0.33
Play time (min)	38.62 ± 2.53	38.52 ± 4.35	0.081	0.936	0.03
Zone 2 (m)	1881.29 ± 225.58	1845.94 ± 362.76	0.348	0.730	0.16
Zone 3 (m)	496.93 ± 103.99	493.13 ± 139.19	0.093	0.927	0.03

Values are mean ± SD; defender group = DG, forward group = FG, distance per minute = DPM, average speed = AVE Speed.

(DG, *n* = 17) and a forward group (FG, *n* = 20) to enable between-group comparisons. Data were collected under standardized conditions to analyze differences in physical characteristics between the groups. All participants were fully informed of the purpose and procedures of the study and voluntarily agreed to participate.

Study Design

This study was designed as a field-based observational study using GPS-derived data to analyze the physical characteristics of young soccer players during match play. Data were collected from two official matches in which the Gyeonggi Province U-17 team participated, and all data were obtained under real competitive conditions. To ensure data consistency and analytical validity, only players who participated for at least 40 min were included in the analysis. This approach minimizes variability due to differences in playing time. Players were classified as forwards and defenders based on their playing positions. All collected data were organized using Microsoft Excel (Microsoft Corporation, Version 2020, Redmond, WA, USA) and subsequently used for the comparative analysis of position-specific physical characteristics.

Measurement Tools

To analyze movement patterns and physical characteristics during matches, a BEE PRO 2 GPS Tracker (SoccerBee GPS Tracker; UBEESLAB Inc., Seoul, Republic of Korea) was used. All participants wore the same GPS device during the matches to ensure consistency. The device was positioned on the upper back of each player to standardize the measurement conditions. Match data were automatically recorded and processed using a dedicated analysis system and then transferred to a computer for further analysis. The primary variables included total distance, HSR, sprinting, and acceleration-deceleration, all of which reflect the external load. All data collection and processing procedures, including device placement and measurement protocols, were standardized to ensure measurement reliability.

Variables

This study analyzed the physical characteristics of young soccer players using GPS-derived data. Based on the previous literature, variables were categorized into five physical components: aerobic capacity, anaerobic capacity, speed and power, neuromuscular function, and agility/change-of-direction (COD). This classification was derived from the biomotor ability framework[12], physiological demands of soccer[13], and recent GPS-based research emphasizing neuromuscular load and acceleration-deceleration[14]. Agility was treated as an independent component, distinguishing it from speed, consistent with a previous study[15]. Aerobic capacity was assessed using indicators of sustained activity, including total distance, distance per minute (DPM), average speed (AVE), playtime, and distance covered in low- to moderate-intensity zones (Zones 2 and 3). Anaerobic capacity was evaluated using indicators of high-intensity performance, including HSR distance (HSR), HSR distance portion (HSR PORTION), distances in zones 4 and 5, number of sprints, sprints per minute (SPM), and top speed. Neuromuscular function was assessed using acceleration- and deceleration-related variables, including acceleration, deceleration, high-intensity acceleration (HI ACCEL), and high-intensity deceleration (HI DECEL). Finally, agility and change in direction were evaluated using COD and agility actions (AGI actions).

Statistical Analysis

All data collected in this study were analyzed using SPSS version 20.0 (IBM Corp., Armonk, NY, USA). Descriptive statistics were calculated and are presented as means and standard deviations. Levene's test was used to assess the homogeneity of variance between the groups. Independent-sample *t*-tests were used to compare differences in physical variables between forwards and defenders. Additionally, Cohen's *d* was calculated to determine the effect

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Table 3. Differences in Anaerobic Capacity

Variables	DG (*n* = 17)	FG (*n* = 20)	*t*	*p*	*d*
HSR Distance (m)	202.93 ± 95.50	251.10 ± 69.33	-1.773	0.085	0.59
HSR Portion (%)	5.06 ± 2.20	6.17 ± 1.60	-1.779	0.084	0.59
Zone 4 (m)	164.64 ± 73.38	192.09 ± 57.60	-1.275	0.211	0.42
Zone 5 (m)	38.29 ± 28.04	59.01 ± 27.84	-2.248	0.031*	-0.74
Sprints (count)	2.71 ± 2.02	3.53 ± 1.67	-1.350	0.185	0.45
SPM (count/min)	0.07 ± 0.05	0.09 ± 0.04	-1.322	0.195	0.44
HPM (count/min)	0.33 ± 0.12	0.38 ± 0.08	-1.427	0.162	0.47
HSR Time (s)	12.59 ± 4.33	14.48 ± 3.45	-1.475	0.075	0.49
Top Speed (km/h)	27.27 ± 1.62	28.39 ± 1.28	-2.350	0.025*	0.78

Values are mean ± SD; defender group = DG, forward group = FG, high-speed running = HSR, sprints per minute = SPM, high-speed runs per minute = HPM.

Table 4. Differences in Neuromuscular Function

Variables	DG (*n* = 17)	FG (*n* = 20)	*t*	*p*	*d*
Acceleration (count)	26.03 ± 18.82	26.25 ± 18.46	-0.036	0.972	-0.01
Deceleration (count)	31.21 ± 5.20	33.23 ± 9.40	-0.788	0.436	-0.26
HI ACCEL (count)	7.29 ± 1.60	8.58 ± 2.59	-1.769	0.086	0.58
HI DECEL (count)	5.44 ± 2.47	5.35 ± 2.87	0.102	0.919	0.03

Values are mean ± SD; defender group = DG, forward group = FG, high-intensity acceleration = HI ACCEL, high-intensity deceleration = HI DECEL.

Table 5. Differences in Agility and Change of Direction

Variables	DG (*n* = 17)	FG (*n* = 20)	*t*	*p*	*d*
COD (count)	8.12 ± 2.15	10.80 ± 4.09	-2.433	0.020*	0.80
AGI Action (count)	81.15 ± 10.89	87.45 ± 17.22	-1.303	0.201	0.43
APM (count/min)	2.10 ± 0.28	2.28 ± 0.37	-1.686	0.101	0.56

Values are mean ± SD; defender group = DG, forward group = FG, change of direction = COD, agility = AGI , actions per minute = APM.

size of between-group differences. The level of statistical significance was set at $p < 0.05$.

RESULTS

The between-group differences in aerobic capacity-related variables are shown in Table 2. Independent-sample *t*-tests revealed no significant differences between the groups in total distance ($t = -0.573$, $p = 0.285$), DPM ($t = -1.060$, $p = 0.148$), AVE speed ($t = -0.996$, $p = 0.163$), play time ($t = 0.081$, $p = 0.468$), Zone 2 ($t = 0.348$, $p = 0.365$), or Zone 3 ($t = 0.093$, $p = 0.463$) (all $p > 0.05$).

The between-group differences in variables related to anaerobic capacity and speed are presented in Table 3. Independent-samples *t*-tests showed significant differences between the groups in HSR ($t = -1.773$, $p = 0.042$), HSR PORTION ($t = -1.779$, $p = 0.042$), Zone 5 ($t = -2.248$, $p = 0.015$), and top speed ($t = -2.350$, $p = 0.012$). In contrast, no significant differences were observed in Zone 4 ($t = -1.275$, $p = 0.105$), number of sprints ($t = -1.350$, $p = 0.093$), SPM ($t = -1.322$, $p = 0.097$), high-speed runs per minute ($t = -1.427$, $p = 0.081$), or HSR time ($t = -1.475$, $p = 0.075$).

The between-group differences in the variables related to neuromuscular function are presented in Table 4. Independent-sample *t*-tests showed a significant difference in HI ACCEL ($t = -1.769$, $p = 0.043$). However, no significant differences were found in acceleration ($t = -0.036$, $p = 0.486$), deceleration ($t = -0.788$, $p = 0.218$), or HI DECEL ($t = 0.102$, $p = 0.459$).

Between-group differences in the variables related to agility and change in direction are presented in Table 5. Independent-sample *t*-tests revealed significant differences in COD ($t = -2.433$, $p = 0.010$) and APM ($t = -1.686$, $p = 0.050$). In contrast, no significant difference was observed in AGI action ($t = -1.303$, $p = 0.101$).

DISCUSSION

This study analyzed match performance characteristics according to playing position in elite young U-17 soccer players. The main findings indicated no statistically significant differences in aerobic-related variables between positions. Nevertheless, compared with defenders, forwards demonstrated higher levels of performance in high-intensity

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and neuromuscular variables, including HSR, top speed, high-intensity acceleration, and COD. These findings suggest that while fundamental aerobic capacity tends to be relatively homogeneous at the elite youth level, high-intensity and non-linear movement demands during match play differ according to positional roles.

The nonsignificant differences in aerobic performance may be attributed to similar training environments and developmental stages among players. Although previous studies have reported positional differences in activity profiles[16,17], these differences may be less pronounced during adolescence owing to the relative homogeneity of physical fitness and the standardization of training intensity[8]. Furthermore, the lack of statistical significance despite the presence of meaningful effect sizes in some variables may have been influenced by the limited sample size and match-related contextual factors.

In contrast, higher performance levels for anaerobic variables, such as HSR and top speed, were observed in forwards. This may be explained by their positional role, which requires frequent high-speed movements such as exploiting space behind defensive lines and performing rapid transitions. Reportedly, forwards and wide players tend to exhibit greater HSR distances and maximal speeds than players in other positions[18,19], reflecting the tactical demands of match play. From an energy metabolism perspective, these high-intensity actions rely heavily on the ATP-phosphocreatine (PCr) energy system because they require rapid power output over short durations[20]. Furthermore, maximal speed performance is influenced by neuromuscular factors, including fast-twitch muscle fiber recruitment, neural activation, and effective utilization of ground reaction forces[21,22].

Similarly, forwards demonstrated higher performance in high-intensity acceleration, which may be associated with repeated demands for rapid increases in speed during match play. In contrast, no significant positional differences were observed in deceleration-related variables, suggesting that deceleration is a fundamental movement requirement common to all positions. Previous studies have also reported greater acceleration performance in forwards and wide players[23,24], and such capabilities have been identified as key indicators of performance levels and future development potential in young athletes[25]. Acceleration requires high levels of neuromuscular activation and rapid force production, and imposes substantial metabolic demands when performed repeatedly[26].

A similar pattern was observed in agility and COD performance. Forwards exhibited higher frequencies and volumes of directional changes, which may be related to repeated movement demands associated with creating space and evading defenders. COD is a fundamental and frequently executed movement in soccer[27], and its frequency and intensity may vary depending on playing position. Previous studies have reported higher COD performance in forwards and wide players[28], which is closely associated with sprint ability and lower limb power[29]. These movements rely heavily on repeated deceleration-reacceleration cycles, eccentric muscle actions, and neuromuscular coordination[30], and their high frequency may also be linked to increased cognitive and decision-making demands during match play.

Despite these findings, this study has several limitations. First, the study comprised a relatively small sample of elite U-17 players, which may limit the statistical power and contribute to the lack of significance in some variables despite the presence of observable effect sizes. Second, because the analysis was restricted to elite players in a single age group, caution should be exercised when generalizing the findings to other age groups or non-elite populations. Third, contextual factors that may influence match performance, such as match outcomes, opponent levels, and tactical strategies, were not fully controlled. Fourth, positions were broadly categorized into forwards and defenders, which may not have fully captured the specific characteristics of the individual subpositions. Finally, the analysis focused on GPS-derived external load variables and did not directly assess internal load or neuromuscular responses, such as muscle activation or physiological markers.

The findings of this study indicate that forwards demonstrated higher levels of performance than defenders in high-intensity and neuromuscular-based variables, including HSR, top speed, high-intensity acceleration, and COD. These results reflect the tactical demands placed on forwards, who are required to perform repeated high-speed movements to exploit space and during rapid transitions. In contrast, no significant differences in aerobic performance were observed between positions, suggesting that baseline physical fitness developed similarly regardless of playing position. Therefore, training programs for forwards should emphasize the enhancement of high-intensity performance capacities such as sprinting, acceleration, and agility. Meanwhile, defenders should focus on maintaining overall physical fitness for sustained match performance while improving deceleration and reactive movement abilities.

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